Report R01 — MGMT Methylation Prediction from Radiomics

Internal validation, UPENN-GBM. Pre-registered null result.

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7 August 2026 • Quantara • Milestone M1 of `EXPERIMENT_PLAN.md`

Summary

We tested whether radiomic features from preoperative multiparametric MRI predict MGMT promoter methylation in glioblastoma. **They do not.** Three model families all performed at or below chance on 256 patients with 1,728 features, and a 200-fold label permutation confirmed the absence of any learnable signal. The pre-specified stopping rule therefore halts the programme at M1: no external validation and no deep-learning work are requested.

The analysis plan, including the stopping threshold, was written and cryptographically hashed *before* any label was read. That hash is embedded in the results file, so the endpoints demonstrably were not chosen after seeing the outcome.

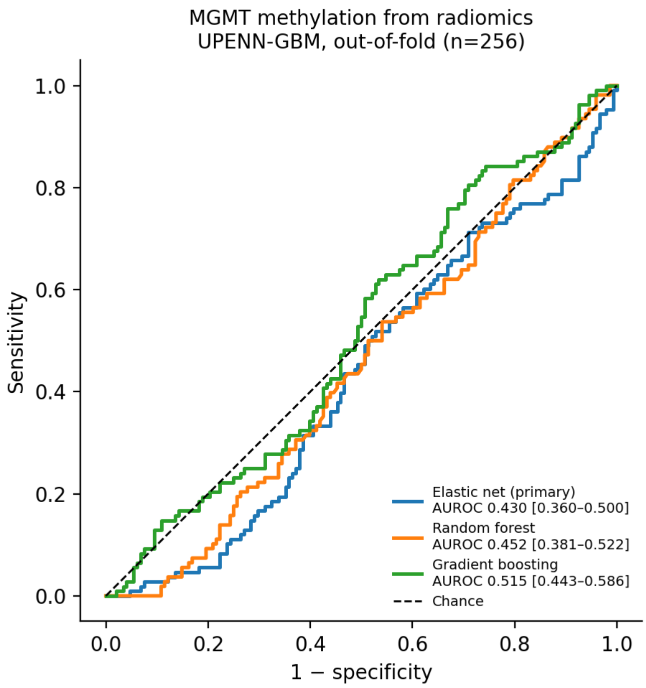
1 What was tested

- **Cohort:** UPENN-GBM (TCIA, CC BY 4.0), development only. The external cohort was never opened in this milestone.
- **Features:** CaPTk radiomic features published with the collection — 4 structural sequences (T1, T1GD, T2, FLAIR) × 3 regions (enhancing tumour, oedema, necrotic core) × 144 features = **1,728 candidates**. Perfusion and diffusion features were excluded *a priori* because the intended external cohort lacks those sequences.
- **Label:** MGMT promoter methylated vs unmethylated, from the collection's clinical file. Indeterminate and unavailable were excluded.
- **Sample: n = 256** complete cases (108 methylated / 148 unmethylated, 42.2% positive).
- **Design:** 10× repeated 5-fold stratified *nested* cross-validation. Feature scaling, variance filtering and elastic-net selection were fitted strictly inside training folds — selection on the full dataset is the most common cause of inflated radiomics performance and was prohibited.
- **Primary model:** elastic-net logistic regression. Random forest and gradient boosting were pre-registered as comparators and are reported regardless of outcome.

2 Result

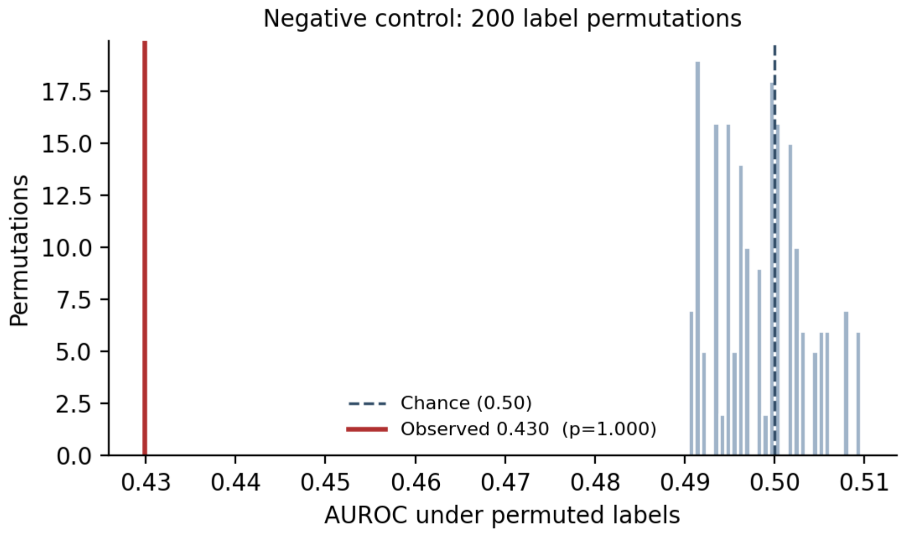
Model	AUROC	95% CI	Per-repeat mean ± SD	Brier
Elastic net *(primary)*	**0.430**	**0.360 – 0.500**	0.459 ± 0.037	0.272
Random forest	0.452	0.381 – 0.522	0.457 ± 0.021	0.254
Gradient boosting	0.515	0.443 – 0.586	0.510 ± 0.027	0.288

Confidence intervals are DeLong intervals on out-of-fold predictions averaged across the ten repeats. No feature exceeded the 10% missingness threshold, so all 1,728 were retained.



Negative control

Two hundred label permutations produced a null AUROC distribution centred on chance (mean 0.498, 95% interval 0.490–0.510). The observed value of 0.430 lies *below* that null, giving $p = 1.000$. This confirms two things: the pipeline contains no leakage (a leaking pipeline shifts the null upward), and the observed performance is not merely weak but indistinguishable from — indeed slightly worse than — random.



3 Decision

The plan specified: *halt if the upper bound of the 95% CI on internal AUROC is below 0.65.* The observed upper bound is **0.500**.

Decision: STOP. M2 (external validation) and M3 (deep learning and habitat analysis) are not justified — there is no signal to validate. No cloud compute was spent on this milestone, and none is requested.

4 Interpretation

This agrees with the literature rather than contradicting it. The RSNA-ASNR-MICCAI BraTS 2021 challenge was built on precisely this task and its organisers concluded the signal may not be reliably learnable from imaging, with leading entries near AUROC 0.6. Our result is a cleanly pre-registered, leakage-controlled replication of that negative finding on an independent cohort.

What this does exclude. That handcrafted radiomic descriptors over automatic segmentations of the four routine structural sequences carry recoverable MGMT information in this cohort at this sample size.

What this does not exclude. A deep-learning model operating on voxels rather than summary descriptors; features derived from perfusion or diffusion sequences (excluded here for transferability); or manual rather than automatic segmentations. We regard the first as substantially less likely given a null this flat across three model families, and we did not test the third: only 59 patients have both manual-segmentation features and a definite label, and running a 59-patient analysis after a null at n=256 would be noise-mining rather than a sensitivity check.

5 Recommendation for the glioma arm

The wider difficulty is label availability, not modelling. Neither cohort in the revised protocol carries a **glioma grade** label at all, and IDH is 19 positives of 630 in development with no external label — so the protocol's stated primary endpoint (grade) is not computable, and IDH is not viable. MGMT was the only endpoint with labels in both cohorts, and it is now tested and null.

We therefore recommend that the glioma arm be re-scoped around an endpoint that has ground truth in an imaged cohort. The two candidates we can see:

1. **Overall survival** — available for all 630 UPENN patients (644 deceased of 671 rows), and a genuine clinical question. No external survival data exists in our holdings, so this would be internally validated only, which should be stated plainly.
2. **An imaged cohort spanning grades 2–4.** UT Southwestern's 625 patients cover the full grade range with IDH, 1p/19q and MGMT, but their images moved behind controlled access during 2025. We hold the molecular labels and not the pixels. Obtaining the images is a request, not a download.

We would value your view on whether either is worth pursuing, or whether the honest conclusion is that this dataset combination cannot support the original question.

6 How to verify this report

Everything in `evidence/` is copied verbatim from an immutable run directory.

- `config.yaml` — the plan as executed. Its SHA-256 (`0ec0a1df...`) is recorded inside `results.json`; a plan altered mid-analysis would not match.
- `gates.json` — eight integrity gates, each with expected value, observed value and PASS/FAIL.
- `oof_*.csv` — the out-of-fold predictions with labels, so every AUROC above can be recomputed independently without rerunning anything.
- `permutation_null.json` — all 200 null AUROC values.
- `results.json`, `inputs.json` (input hashes), `env.txt` (package versions), `MANIFEST.sha256`, `m1_internal.py` (the code).

- gates_FAILED_first_attempt.json — see below.

Two disclosures we would rather make than omit

A gate caught an error of ours before any model was fit. The first attempt halted at the cohort-size gate: expected 256, observed 262. The twelve feature files do not cover identical subjects (enhancing tumour 604, oedema 611, necrotic core 602), and our planning figure had been derived from a single file. Taking the union of subjects would have median-imputed entire 144-feature blocks for four patients — fabricating data. We switched to complete cases across all twelve files, giving n=256, and documented the superseded figure in the source. The failed run is retained rather than deleted, and its gate record is included as gates_FAILED_first_attempt.json.

Determinism was tested, not assumed. An independent re-run reproduced byte-identical predictions, and a second full run performed after adding prediction output reproduced all three AUROCs to twelve decimal places.

Run directory: paper-draft/02-revised-protocol-open-datasets/runs/20260806T223856Z-3f8bece. Data: UPENN-GBM, TCIA, CC BY 4.0 (Bakas et al., DOI 10.7937/TCIA.709X-DN49). No patient-level data is reproduced in this report.